

TEFAR — TMS-EEG / EEG FieldTrip Artefact Removal

A lightweight, configurable framework for ICA-based artefact-component classification in (TMS-)EEG. One scoring engine, two profiles. FieldTrip is the only dependency — no MATLAB toolboxes (kurtosis, robust statistics, autocorrelation and the periodogram used in the validation are all implemented locally).

Files

File	Purpose
<code>tefar_core.m</code>	The engine. Runs ICA (FastICA by default) and scores every component against a configurable set of detectors.
<code>TEFAR_tms.m</code>	TMS profile wrapper. Drop-in replacement for TEFAR_v3 — first seven outputs are identical, so existing pipelines keep working.
<code>TEFAR_eeg.m</code>	EEG profile wrapper for ordinary resting/task data (no TMS).
<code>simulate_tefar_data.m</code>	Ground-truth simulation ($x = A \cdot s$) with known artefact labels. Base MATLAB/Octave only.
<code>validate_tefar.m</code>	Validation against the simulation using the real <code>tefar_core</code> (needs FieldTrip).
<code>verify_tefar_logic.m</code>	Independent, FieldTrip-free reimplementations of the detector math for a toolbox-free cross-check (runs in MATLAB or Octave).
<code>_verify_logic.py</code>	Python port of the simulation + detector logic used during development.

Detectors

Each detector targets an established independent-component artefact signature:

Detector	Feature	Rule
line	mains fundamental / sideband prominence	robust-z > 5
muscle	high-frequency power fraction (mains excluded)	robust-z > 5 and fraction > 0.5
muscle_topo	spatial focality = peak/ Σ topo	robust-z > 5
blink	temporal kurtosis + frontal topography	kurt > 4 and frontal ratio > 0.5
eyemove	fronto-lateral anti-symmetry	asym > 0.15 and frontal ratio > 0.5
cardiac	autocorrelation periodicity + spikiness	robust-z > 5, autocorr > 0.15, kurt > 3.5
decay (TMS)	early post-pulse RMS vs baseline SD	RMS > 2·baseline SD
recharge (TMS)	later-window RMS vs baseline SD	RMS > 4·baseline SD

Two design choices carried over from testing:

- **Robust z-scores (median / MAD)** replace `mean + k·SD`. With a single dominant artefact component the classic rule is nearly unreachable (the outlier inflates its own mean and SD; the maximum achievable z is $(N-1)/\sqrt{N}$). Median/MAD is immune to this.
- All temporal detectors locate their window from `comp.time` **per trial**, so there is no assumption about trial/segment structure (the original `TEFAR_v3` hard-coded a 3-segments-per-trial layout).

Usage

```
% TMS-EEG, drop-in for the old call (existing pipeline unchanged):
[comp, line_c, musc_c, decay_c, addmusc_c, rech_c, blink_c] = ...
    TEFAR_tms(data_filtered, trl);

% ...or grab the full result (scores, metrics, populated cfg):
[comp, ~,~,~,~,~,~, artifacts] = TEFAR_tms(data_filtered, trl);
artifacts.reject           % suggested components
```

```

artifacts.score      % detectors firing per component
artifacts.metrics    % all per-component metric vectors

% Ordinary EEG:
[comp, artifacts] = TEFAR_eeg(data);

% Override any default (e.g. US mains, stricter line rule, score-
based reject):
cfg =
struct('line_freq',60,'reject_rule','score','reject_threshold',2);
cfg.line.z = 6;
[comp, artifacts] = TEFAR_eeg(data, cfg);

```

`reject_rule` is `'union'` by default (any detector flags → suggest removal), matching the original behaviour. Set `'score'` to require agreement across `reject_threshold` detectors instead. Either way, TEFAR only **suggests** — confirm by visual inspection before rejecting.

Validation

$x = A \cdot s$ is built from labelled latent sources, so every flag can be scored against ground truth. Feeding the known sources straight into the detectors (isolating classification from ICA separation quality):

Profile	Sensitivity	Specificity	Accuracy
TMS (single seed)	1.00	1.00	1.00
EEG (single seed)	1.00	1.00	1.00
TMS (40 random seeds)	~0.93	~1.00	~0.98
EEG (40 random seeds)	~0.92	~0.99	~0.97

Near-perfect **specificity** is the property that matters most: brain components are essentially never removed. The ~8 % of missed artefacts on some seeds are exactly what the semi-automatic visual-inspection step exists to catch. Reproduce with:

```

verify_tefar_logic('tms');    verify_tefar_logic('eeg');    % toolbox-
free
validate_tefar('tms','components');    % via real

```

```
tefar_core
validate_tefar('tms', 'endtoend');           % full ICA
pipeline
```

Note on porting from TEFAR_v3

The refactor also fixes a few latent issues in the original: `kurtosis()` required the Statistics toolbox (now local); the recharge window `-1.5...1.5 s` selected the whole segment; decay used signed-mean (sign cancellation) rather than RMS; and the multi-metric `score` was computed but never actually used for rejection (the pipeline took the union of all categories). All are addressed here.